9 **(a)** State the *Principle of Superposition*.

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.....

..... [2]

(b) Laser beam of red light of wavelength 644 nm is incident normally on a diffraction grating having 550 lines per millimetre, as illustrated in Fig. 9.1.

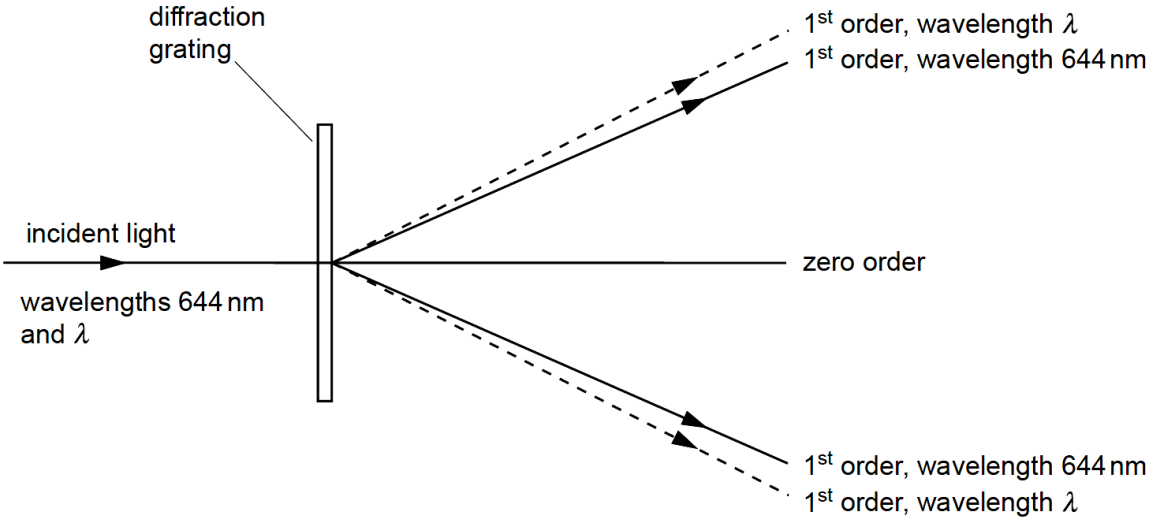

Fig. 9.1

Another laser beam of red light of wavelength λ is also incident normally on the grating. The first order diffracted light of both wavelengths is illustrated in Fig. 9.1.

(i) Determine the total number of bright spots of wavelength 644 nm that are visible.

total number of bright spots = ... [3]

(ii) State and explain

1. whether λ is greater or smaller than 644 nm,

.....

.....

..... [1]

2. in which order of diffracted light there is the greatest separation of the two wavelengths.

.....

.....

..... [1]

(c) The laser beam of red light of wavelength 644 nm is now placed in front of two slits as shown in Fig. 9.2.

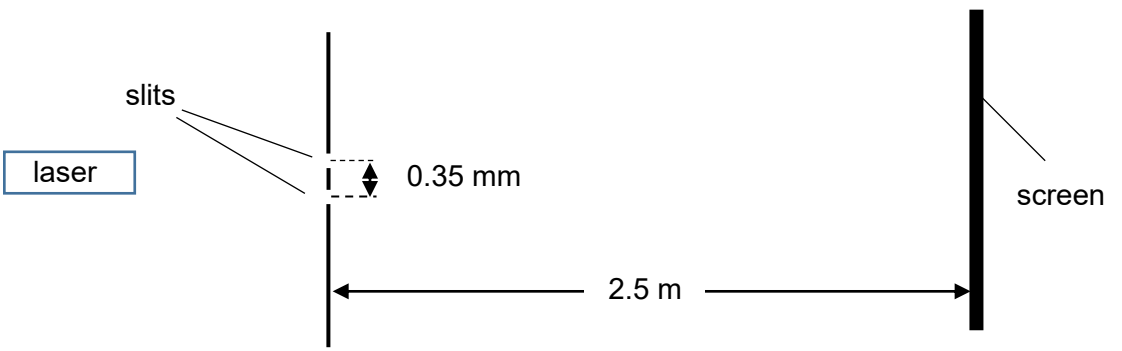

Fig. 9.2 (not to scale)

The distance from the slits to the screen is 2.5 m. The separation of the slits is 0.35 mm. The width of each slit is 2.4×10^{-5} m. The maximum intensity of the interference pattern formed on the screen is I.

(i) Explain how the interference pattern is formed on the screen.

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..... [3]

(ii) One of the slits is now covered with an opaque object.

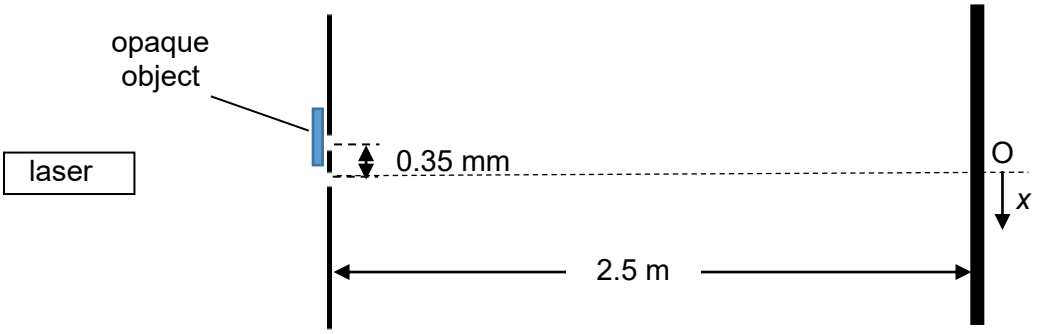

Fig. 9.3

An interference pattern is observed on the screen centred at O.

1. Calculate the width of the central fringe observed on the screen.

width = ... m [3]

2. On Fig. 9.4, sketch a graph to show the variation with distance *x* from point O of the intensity of the light observed on the screen. Draw at least 3 maxima.

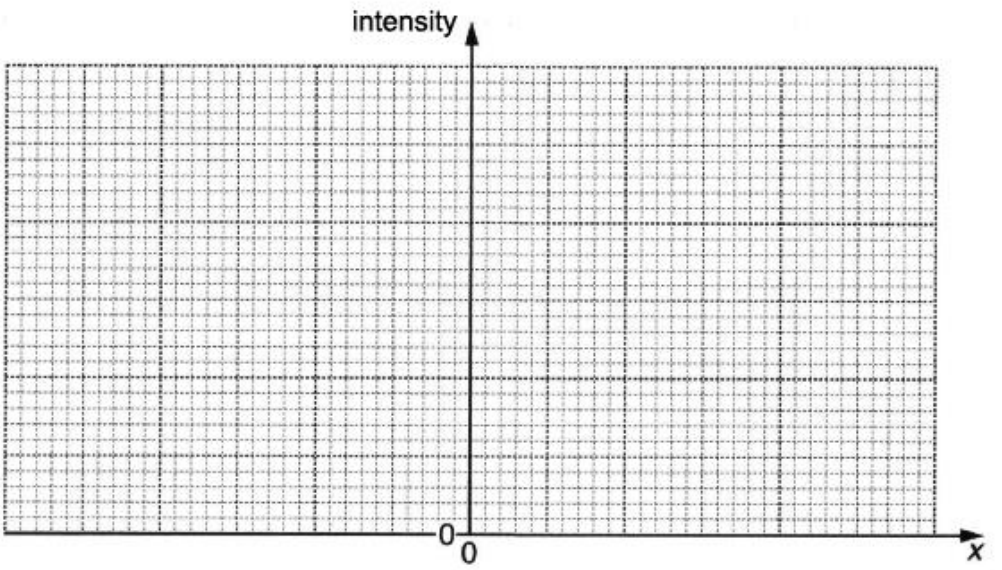

Fig. 9.4

[3]

3. Deduce, in terms of I, the maximum intensity of the interference pattern produced.

intensity = ... [2]

(iii) The laser together with another identical laser are positioned as seen in Fig. 9.5 such that the light from both sources pass through the uncovered slit at an angle of θ to each other.

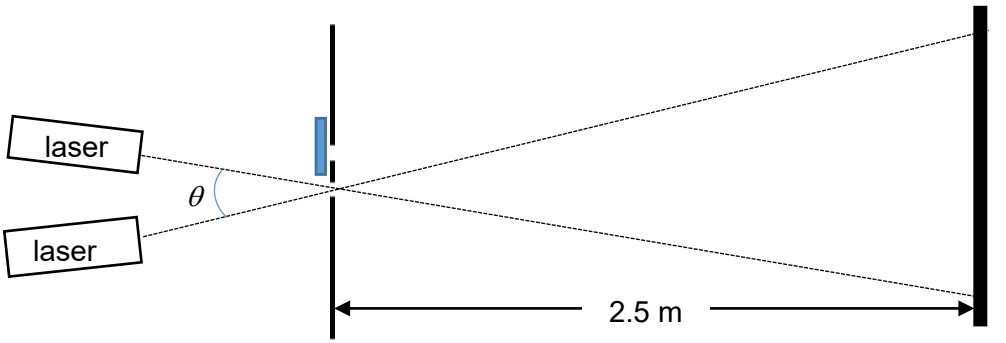

Fig. 9.5

State and explain whether the interference patterns formed by the two sources are resolved if θ is equal to 2.0°.

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..... [2]